

# ABDUL MANAN

☎ +92 335 5100705 | ✉ abdulmananshah02@gmail.com | in abdul-manan02 | 🌐 abdulmanan02

## ABOUT ME

---

Software Engineer with proven experience delivering scalable solutions across FinTech, Retail, AI, and Industrial Automation. Skilled in Full-Stack Web Development, AI Automations, and DevOps. Excels within multi-disciplinary teams, engaging directly with clients to interpret complex business needs into robust technical solutions that drive organisational impact.

## EXPERIENCE

---

### Software Engineer (Backend)

Sept 2025 - Current

Virtual World (IBEX)

*Remote*

- Designed & implemented a **Multi-Tenant Architecture** for a Ticketing System using Nest.js, PostgreSQL, and Keycloak, ensuring database isolation per tenant and enabling scalable client onboarding for major clients including FIA, Jazz, and Shifa.
- Optimized API performance by replacing sequential database calls with parallel execution and integrating data loaders to cache and batch related queries, **reducing queries by over 80%**.
- Built the entire **Workflows module** to enable ticket escalations to relevant stages, departments, and users while integrating multiple communication channels.
- Designed a custom **Audit Logging** system with a dedicated change log table to capture all data mutations, persisting before/after state snapshots along with actor metadata for full operational traceability.
- Designed a custom **Audit Logging** system with a dedicated change log table to capture all data mutations, persisting before/after state snapshots along with actor metadata for full operational traceability.

### Polyglot Engineer (Full Stack & DevOps)

June 2024 - July 2025

Cowlar Design Studio (YC-17)

*Pakistan*

- Engaged directly with clients to understand pain points, propose solutions, and refine them through iterative feedback. Led projects from 0-to-1, **owning the full Software Development Lifecycle (SDLC)** from requirements and UI design to team execution, client handoff, and post-launch support, all while serving as a dedicated Full-Stack and DevOps resource.
- Led the end-to-end development of a Subscription Management Project for a Qatari **FinTech** company, translating client needs into actionable goals and managing the execution of the development team.
- Optimized MongoDB queries, reducing query time by 75% on datasets with **17 million+ records**, enhancing data retrieval performance.
- Built a Gift Card Processing System, managing **QAR 4 million+** in transactions within 20 days of launch, ensuring system scalability and reliability.
- Designed & developed a Web Application using AWS Amplify, Lambda, and S3 to manage **thousands of simultaneous file uploads**, replacing a manual, error-prone process and improving visibility.
- Collaborated on developing annotation tools and integrating in-house and **Azure Machine Learning models**, streamlining the annotation process and cutting annotation time by over 50%.
- Worked on multiple products within our **Retail Ecosystem** such as E-Commerce Applications along with Customer and Admin Portals.
- Architected a **Bare-Metal Kubernetes Cluster** with MetalLB for load balancing, Calico for network policies, CRI-O as the container runtime, and Helm for app management, ensuring scalability and efficiency.
- Developed **CI/CD Pipelines** on GitLab/GitHub for seamless continuous integration and deployment.
- Leveraged **Node.js Worker Threads** to offload CPU-intensive tasks from the main event loop, improving application throughput and reducing response latency under high concurrency.
- Integrated various third-party features, including Email, WhatsApp and SMS along with **Apple, Samsung, and Google Wallet**.

- Implemented **MQTT protocol** for real-time IoT communication, using it as a broker for efficient data transmission between devices and applications.
- Built & maintained a **custom Role-Based Access Control (RBAC)** system to manage fine-grained user permissions and enhance system security.

Full Stack Developer

February 2024 - April 2024

ezWifi (Techstars 23')Remote

- Developed an **analytics dashboard** to monitor and visualize key metrics for deployed ezRouters, delivering actionable insights to clients.
- Enhanced an **admin portal** to support remote router management, enabling administrators to perform configuration changes and restarts remotely.

Full Stack Developer

December 2023 - April 2024

FreelancingRemote

- Stabilized **E-Commerce applications** by fixing bugs and developing new features as required.
- Worked with **Stripe API** to handle payments and subscriptions.

SKILLS

|                  |                                                                       |
|------------------|-----------------------------------------------------------------------|
| Frontend         | Vue.js, React.js                                                      |
| Backend          | Node.js, Nest.js, Express, Fastify, Python, FastAPI                   |
| Databases & ORMS | PostgreSQL, MongoDB, IndexedDB, InfluxDB, Sequelize, Prisma, MikroORM |
| CI/CD Tools      | GitHub Actions, GitLab CI/CD                                          |
| Cloud Platforms  | Amazon Web Services (AWS)                                             |
| DevOps           | Git, Docker, Kubernetes, Helm                                         |
| Others           | GraphQL, Swagger, MQTT, Protocol Buffers, Linux, Bash Scripting       |
| Languages        | JavaScript, TypeScript, MongoDB Query Language, SQL                   |

EDUCATION

|                                         |                     |
|-----------------------------------------|---------------------|
| Fast Nuces                              | Islamabad, Pakistan |
| Bachelor of Science in Computer Science | 2020-2024           |
| Supernova                               | Islamabad, Pakistan |
| Cambridge International A-Levels        | 2018-2020           |